DYLAN J. WOLFF

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EDUCATION

National University of Singapore 2021 - 2026
Ph.D. in Computer Science (GPA: 5/5) *Singapore, SG*
Thesis: *Vertical and Horizontal Expansion of Fuzzing Techniques*

ETH Zürich 2018 - 2020
M.S. in Computer Science, Concentration: Information Security (GPA: 5.33/6) *Zürich, CH*
Thesis: *Value Mutation Testing for SMT Solvers*

Boston College 2011 - 2015
B.S. in Computer Science, Minor in Mathematics (GPA: 3.79/4, *magna cum laude, honors program*) *Newton, MA*
Thesis: *Mutational Fuzzing to Discover Software Vulnerabilities*

PUBLICATIONS

1. [**USENIX SEC.'26**] Jiacheng Xu*, Dylan Wolff*, Xing Yi Han, Jialin Li, Abhik Roychoudhury. *"Concurrency Fuzzing of the Linux Kernel with eBPF"* 35th Usenix Security Symposium, 2026
2. [**S&P'26**] Zhengxiong Luo, Huan Zhao, Dylan Wolff, Cristian Cadar, Abhik Roychoudhury. *"Agentic Concolic Execution"* 47th IEEE Symposium on Security and Privacy, 2026
3. [**CACM'25**] Dylan Wolff, Martin Mirchev, Abhik Roychoudhury. *"Large Language Models in Software Security Analysis."* Communications of the ACM (CACM), 2026
4. [**EMSE'25**] Dylan Wolff, Yannic Noller, Ridwan Shariffdeen, Abhik Roychoudhury. *"Shifting Fuzzing Left in Software Workflows."* Empirical Software Engineering (EMSE), 2025
5. [**TOSEM '25**] Dylan Wolff, Marcel Böhme, Abhik Roychoudhury. *"Fuzzing: On Benchmarking Outcome as a Function of Benchmark Properties."* Transactions on Software Engineering and Methodology (TOSEM), 2025
6. [**ASPLOS '25**] Huan Zhao, Dylan Wolff, Umang Mathur, Abhik Roychoudhury. *"Selectively Uniform Concurrency Testing."* 30th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2025
7. [**ASPLOS '24**] Dylan Wolff, Zheng Shi, Gregory J. Duck, Umang Mathur, Abhik Roychoudhury. *"Greybox Fuzzing for Concurrency Testing."* 29th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), 2024

WORK EXPERIENCE

Amazon Web Services 2026
Applied Scientist *Boston, MA*

Amazon Web Services 2025
Applied Scientist Intern *Seattle, WA*

- Optimized the performance of Shuttle, an open-source Rust concurrency testing tool, including novel yield-point optimizations; reduced runtime by 50% on large-scale storage applications within S3
- Introduced notion of "time models", enabling Shuttle to explore timing related concurrent behaviors such as async timeout cancellation
- Refactored Shuttle yield-point placement and added stable task and resource identifiers to facilitate advanced scheduling algorithms

Mathworks 2015-2018
Support and Software Engineer *Natick, MA*

- Developed custom internal testing and deployment tools, primarily in Java, to facilitate the infrastructure shift towards Docker containers orchestrated through Kubernetes for web applications such as MATLAB Online
- Rotated through several software development projects on different teams across the company ranging from C++ development of core MATLAB to Java/JS centric web infrastructure
- Received only $\geq 4/5$ in overall customer satisfaction surveys over the course of 13.5 months of technical phone and email support of MATLAB ($\mu = 4.8$); selected as a support team leader for handling escalations and organizing shifts

ACADEMIC AWARDS AND HONORS

President's Graduate Fellowship (NUS) 2021
John J. Neuhauser Award (BC) – Awarded annually for most outstanding achievement in Computer Science 2015

TECHNICAL SKILLS

Languages	Rust, Python, Zig, C, Java, MATLAB, Javascript, SQL, Datalog, Viper, SMT-LIB
Technologies	LLVM, Docker, Pandas, Z3, Maven, Kubernetes, eBPF, Verus
Other Skills	Fuzzing, Symbolic Execution, Program Analysis, Deductive Verification, Reverse Engineering

ACADEMIC SERVICE

IEEE/ACM International Conference on Software Engineering (ICSE) – *Shadow PC Member* 2026
IEEE Transactions on Software Engineering (TSE) – *Reviewer* 2026
ACM Transactions on Software Engineering and Methodology (TOSEM) – *Reviewer* 2025 - 2026
SOSP Artifact Evaluation – *Program Committee* 2025
ACM International Conference on the Foundations of Software Engineering (ESEC/FSE) – *Student Volunteer* 2022

TEACHING

Fuzzing Summer School – *Lecturer and Competition Organizer* 2024,2026
Advanced Topics in Program Analysis – *Project Mentor* 2024
Automated Software Validation – *Teaching Assistant* 2022,2023

MENTORSHIP

Xing Yi Han – **Masters Thesis:** Custom Kernel Scheduler With eBPF For Concurrency Fuzzing 2024
Dick Ong and Ivan Tan – **Final Year Undergrad. Project:** Python Concurrency Testing 2025 - 2026
Zi Shen Lim, Charles Wen – **Semester Undergrad. Project:** Multi-Process Python Concurrency Testing 2026 - Present
Ian Cheah, Luoqi Xu – **Final Year Undergrad. Project:** Dev. X for Python Concurrency Testing 2026 - Present

SELECTED PROJECTS

DARPA AIxCC 2023-2024
Team Lead (International Cybersecurity Competition) *NUS*

- Coordinated an international team across 4 universities in creating a system which can autonomously find and fix software vulnerabilities – combines static analysis, fuzzing, localization, repair and validation into a single pipeline; team eliminated in semi-final
- Implemented several components such as a static program-slicing statistical fix localization tool using binary instrumentation and a multi-lingual fault injection tool in Python to generate training data for learning-based repair methods

FUN FACT

Former semi-professional ultimate frisbee player; member of the World-Champion USA National U23 team in 2013